



C2 Cell respiration

C · Interaction and interdependence · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Cell respiration is the controlled release of chemical energy from organic molecules to produce ATP. Aerobic respiration begins with glycolysis in the cytoplasm (glucose → 2 pyruvate, net yield 2 ATP), followed by the Krebs cycle and oxidative phosphorylation in the mitochondria, where oxygen acts as the final electron acceptor, together yielding roughly 30 or more ATP per glucose. Without oxygen, anaerobic respiration (fermentation) regenerates NAD⁺ so glycolysis can continue, producing lactate in animal cells or ethanol and CO₂ in yeast, but with a much lower net yield of only 2 ATP per glucose.

Worked example: follow the reasoning

Comparing ATP yield of aerobic versus anaerobic respiration of the same amount of glucose. Step 1: aerobic respiration of 1 mole of glucose yields approximately 30 ATP (glycolysis 2 net + Krebs/oxidative phosphorylation the rest). Step 2: anaerobic respiration (lactic acid fermentation) of 1 mole of glucose yields only 2 ATP net, from glycolysis alone. Step 3: ratio of ATP yield = $30 \div 2 = 15$. Conclusion: aerobic respiration extracts about 15 times more ATP from the same amount of glucose than anaerobic respiration, explaining why sustained activity requires oxygen while intense short bursts rely on the far less efficient anaerobic pathway and accumulate an oxygen debt.

Quick check · multiple choice

In which part of a eukaryotic cell does the Krebs cycle take place?

- A Cytoplasm
- B Mitochondrial matrix
- C Inner mitochondrial membrane
- D Nucleus

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A respirometer was used to measure the rate of oxygen consumption of germinating pea seeds at 15°C, 25°C and 35°C, giving oxygen uptake rates of 0.4, 0.9 and 1.6 mm³ min⁻¹ respectively. [7]
 - (a) Describe the trend shown by this data and suggest a biological explanation. [3]
 - (b) Explain why oxygen consumption is a valid indicator of the rate of aerobic cell respiration. [2]
 - (c) Predict, with a reason, what would happen to the measured rate if the experiment were repeated at 50°C. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) Oxygen consumption increases with temperature across the range tested (0.4 to 1.6 mm³ min⁻¹); this is because higher temperature increases the kinetic energy of enzyme and substrate molecules involved in respiration, increasing the frequency of successful collisions and therefore the rate of the enzyme-catalysed reactions of aerobic respiration.
- (b) Oxygen is consumed specifically as the final electron acceptor in oxidative phosphorylation during aerobic respiration, so the rate at which it is used reflects the rate at which the electron transport chain (and therefore aerobic respiration overall) is operating.
- (c) The rate would likely be lower than at 35°C (or than expected from the trend), because 50°C is likely above the optimum temperature for the respiratory enzymes, causing denaturation of enzyme tertiary structure and a fall in the rate of the enzyme-catalysed reactions of respiration.

Quick check answer

B. The Krebs cycle enzymes are located in the mitochondrial matrix, where pyruvate-derived acetyl groups are fully oxidised, releasing CO₂ and generating reduced electron carriers (NADH, FADH₂).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

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